CHAPTER

16

Designing Interaction

Highlights

- Designing for interaction needs
- Creating an interaction design
- Storyboards
- Wireframes
- Intermediate interaction design
- Interaction design production
- Design systems

16.1 INTRODUCTION

16.1.1 You Are Here

We begin each process chapter with a "You are here" picture of the chapter topic in the context of The Wheel, the overall UX design lifecycle template (Fig. 16.1). In this chapter, we describe how to design for the middle layer of the human needs pyramid—the interaction.

In Section 11.2.1, we discussed how interaction needs are about being able to manipulate the user interface controls to perform required tasks in the work domain using the product or system being designed. In this chapter, we go about designing the interaction to satisfy those user needs.

16.2 DESIGNING FOR INTERACTION NEEDS

16.2.1 Designing for Interaction Needs Is About Supporting User Tasks

In practical terms, interaction design is about how people use the system or product to perform tasks within the broader work practice and covers all touch points where the user interacts with the ecology. It is where users look at displays and manipulate controls, doing sensory, cognitive, and physical actions. Examples of interaction design in the Apple Music ecology include supporting users with signing up for an Apple ID;

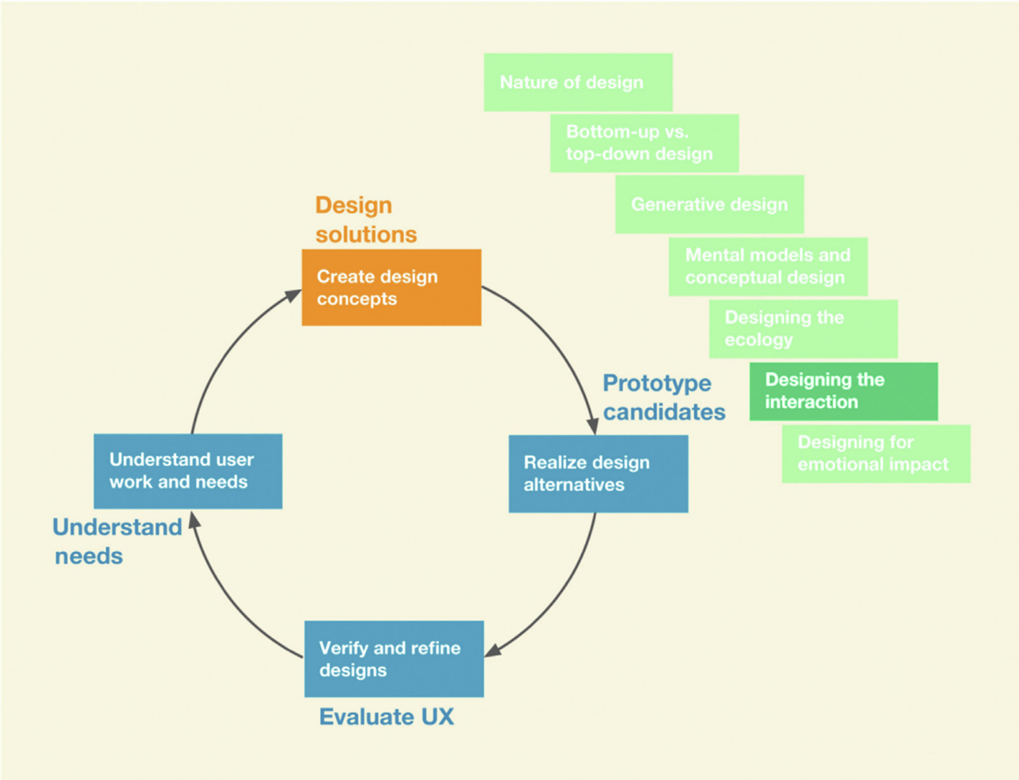

Fig. 16.1

You are here in the chapter describing designing the interaction, within the design solutions activity, in the context of the overall Wheel UX lifecycle.

logging in; searching or browsing for songs by artist, title, album, and rating; using voice assistant Siri or Apple Music for a variety of tasks, such as selecting, playing, pausing, and rating songs; buying Apple Music subscriptions; and offering curation support for users to build their own playlists.

16.2.2 Different Device Types in the Ecology Require Different Interaction Designs

Different devices have different form factors, usage conventions, constraints, and capabilities, each requiring an appropriate design for interaction with that device. For example, the interaction design for a search task will be different for desktops, phones, tablets, and watches—with smaller and less capable devices providing fewer options and more constrained results. Even within the same device category, such as a phone, interaction designs will differ to accommodate the various target platform conventions. For example, the interaction design of an application on the Android platform will be different from that of the Apple platform.

16.3 CREATING AN INTERACTION DESIGN

16.3.1 First Identify All Devices and Their Roles in the Ecology

With your UX team, begin by enumerating all the devices that are envisioned for the ecology. What role will each of these devices play? For example, depending on the nature of work practice, you can think of a smartwatch as mostly a tracking and notification device, whereas a desktop can be the central hub where the majority of the content is generated. Other devices, such as phones and tablets, can be used for the combination of both content generation and consumption tasks.

16.3.2 Proceed With Generative Design

Get immersed in the task-related models, requirements, and user stories, and involve the team in ideation and sketching using the techniques discussed in Chapter 13. The first goal is to create as many ideas as possible for the overall theme or metaphor of the interaction conceptual design (Section 16.3) for each device in the ecology. Then switch to generating ideas for interaction patterns and how the structure of the dialogue between user and system will proceed for each device. Think of how a given task sequence can be handled by multiple devices as the user switches contexts in the ecology. After the ideas are generated and captured as sketches, the team will critique each idea and concept for tradeoffs.

16.3.3 Establish a Good Conceptual Design for the Interaction

A conceptual design (see Sections 14.2 and 14.3) for the interaction can be based on a design pattern (repeatable solution to a common design problem that emerges as best practice, encouraging sharing and reuse) or a metaphor to describe how the interaction design works for a given device.

An example is a calendar application on a desktop in which user actions look and behave like writing on a real calendar. A more modern example is the metaphor of reading a book on an iPad. As the user moves a finger across the display to push the page aside, the display takes on the appearance of a real paper page turning.

Also recall from Section 14.3.6 the example of using a time machine metaphor in the conceptual design of a backup feature available in Apple's Mac operating system. That metaphor explains the somewhat technical and cumbersome activity of data backup and recovery to everyday users.

16.3.4 Leverage Interaction Design Patterns

Design patterns (Borchers, 2001; Tidwell, 2011; Welie & Hallvard, 2000) are repeatable solutions to common design problems that emerge as best practices, encouraging sharing and reuse and promoting consistency. Much of what we see in modern graphical user interfaces and mobile interfaces today is built up of design patterns (e.g., the way a button looks and acts or overall behavior of a search feature).

A pattern library (Schleifer, 2008) is like a beefed-up design system (Section 16.9). Like design systems, they help you avoid having to reinvent designs for common repetitive local design situations within the greater scope of interaction designs. They are a way of sharing design experience about widget-level design that has been evaluated, refined, and proven to be workable.

Leverage design patterns, concepts from the work domain, interaction ideas, and workflows for each device. Some domains and products have established patterns that can be embraced. For example, suppose you are working on an email communication system. If you want to take advantage of an established standard interaction design concept that we know works for desktop systems, then you will feature a three-way split pane pattern with a list of mailboxes as one pane, a list of messages of the selected mailbox in the second pane, and the preview of the selected message in the third pane. However, if the goal is to set yourself apart from the crowded space of email systems, then you will break with the established pattern and come up with a better and more innovative design concept.

Another example of using a concept or idea from the target work domain in an interaction design is the use of a shopping cart on an online shopping website. This pattern communicates how the design works in the digital realm. As users shop, they can click on the cart icon to see what is in their cart, just like shopping in the physical world.

16.3.5 Establish the Information Architecture for Each Device

In Chapter 15, we talked about establishing the pervasive information architecture (which provides ever-present information availability across devices and users) for the entire ecology. Now, for each device, you also need to decide on the information architecture. What information will be available for interaction on each device? How will it be structured? What happens when a user tries to access information not available on that device? What is the best way to represent that information in the design? What are the modalities with which users can access that information (voice, touch, etc.)?

Example: Interaction Conceptual Design for the Ticket Kiosk System

There is a common perception of a ticket kiosk that includes a box on a pedestal and a touchscreen with colorful displays showing choices of events. If you give an assignment to a team of students, even most UX students, to come up with a conceptual design of a ticket kiosk in 30 minutes, 90% of the time you are likely to get something like this. If you teach students to approach it with ideation and sketching, however, they can come up with amazingly creative and varied results.

In our ideation about the Ticket Kiosk System, someone mentioned making it an immersive experience. That triggered more ideas and sketches on how to make it immersive, until we came up with a three-panel overall design that literally surrounds and immerses the user (Fig. 16.2).

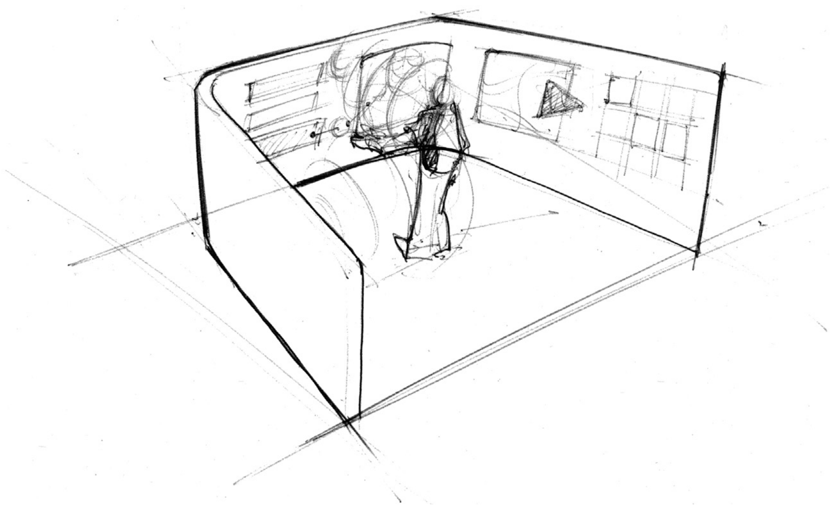

Fig. 16.2

Part of a conceptual design showing immersion in the emotional perspective (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

Following is a brief description of the concept, written by the UX team in outline form.

- The center screen is the interaction area, where immersion and ticket-buying action occur.
- The left screen contains available options or possible next steps (e.g., this screen may provide a listing of all required steps to complete a transaction, including letting the user access these steps out of sequence).

- The right screen contains contextual support, such as interaction history and related actions (e.g., this screen may provide a summary of the current transaction so far and related information, such as reviews and ratings).
- Each next step selection from the left panel puts the user in a new kind of immersion in the center screen, and the previous immersion situation becomes part of the interaction history on the right panel.
- To address privacy and enhance the impression of immersion, when the ticket buyer steps in, rounded shields made of classy materials gently wrap around.
- An “Occupied” sign glows on the outside.
- The inside of the two rounded half-shells of the shield becomes the left and right interaction panels.
- Note: We may need to evaluate whether this could induce feelings of claustrophobia.

Fig. 16.3 shows part of a more traditional concept for the Ticket Kiosk System interaction design. The available categories of tickets are displayed as a menu, as the starting point. When users select a category, they go to a screen for that category, and a curated set of featured events is displayed in a list. The details of navigating among these choices and accessing the full list

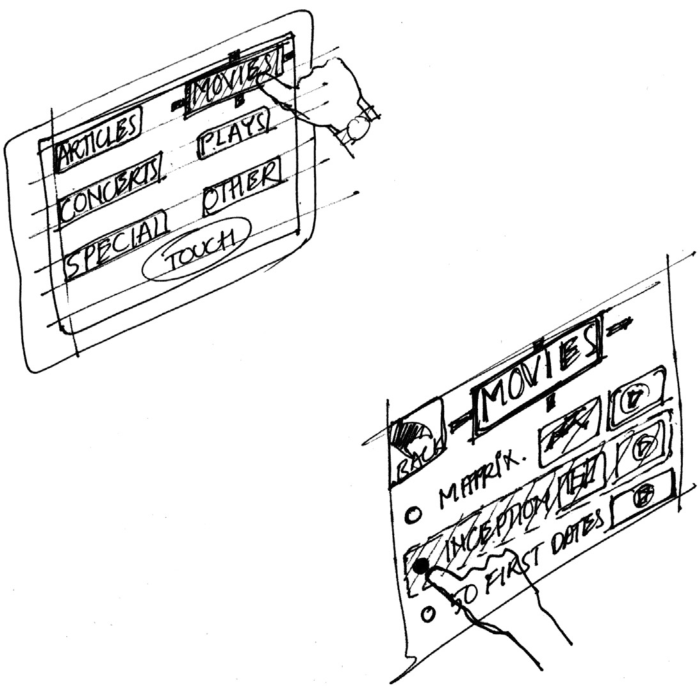

Fig. 16.3

Part of a conceptual design in the interaction perspective (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

of available events in this category are to be fleshed out in further rounds of ideation.

Exercise 16.1 Conceptual Design for Interaction in Your Product or System

Think about UX research data for your system and envision a conceptual design for interaction (see Section 5.6.1.2). Try to communicate the designer's mental model of how the user operates the system, perhaps along the lines of how we did in the previous example.

16.3.6 Envision Interaction Flows Across Different Devices in the Ecology

Even though we design interactions based on device type, users do not think about their work that way. For users, work gets done in an ecology. So, it is important to support the overarching goals of the users as they transition from one device to another. For example, a user's high-level goal of buying a product may involve searching for it on a desktop using a website, placing an order, getting an email confirmation with details on when it will be available for pickup, getting a text message to notify that it is ready for pickup at a local store, and receiving a confirmation message to show in the store during item pickup.

It is important for interaction designers to keep in mind this high-level user goal of buying a product as they go about designing for each individual device. In the e-commerce example, designing interaction on the smartphone may include a feature to use location awareness capability to show the order confirmation when the user is detected to be at one of the company's stores. So, when the user opens the application on the phone to pull up the confirmation, the system anticipates the need and shows the "latest order" information by default.

One way to think about and model such work goals is through the use of storyboards, which we discuss next.

16.4 EXAMPLE OF STORYBOARD FOR THE INTERACTION PERSPECTIVE

Fig. 16.4 shows sample storyboard sketches for buying a concert ticket using a three-screen kiosk design.

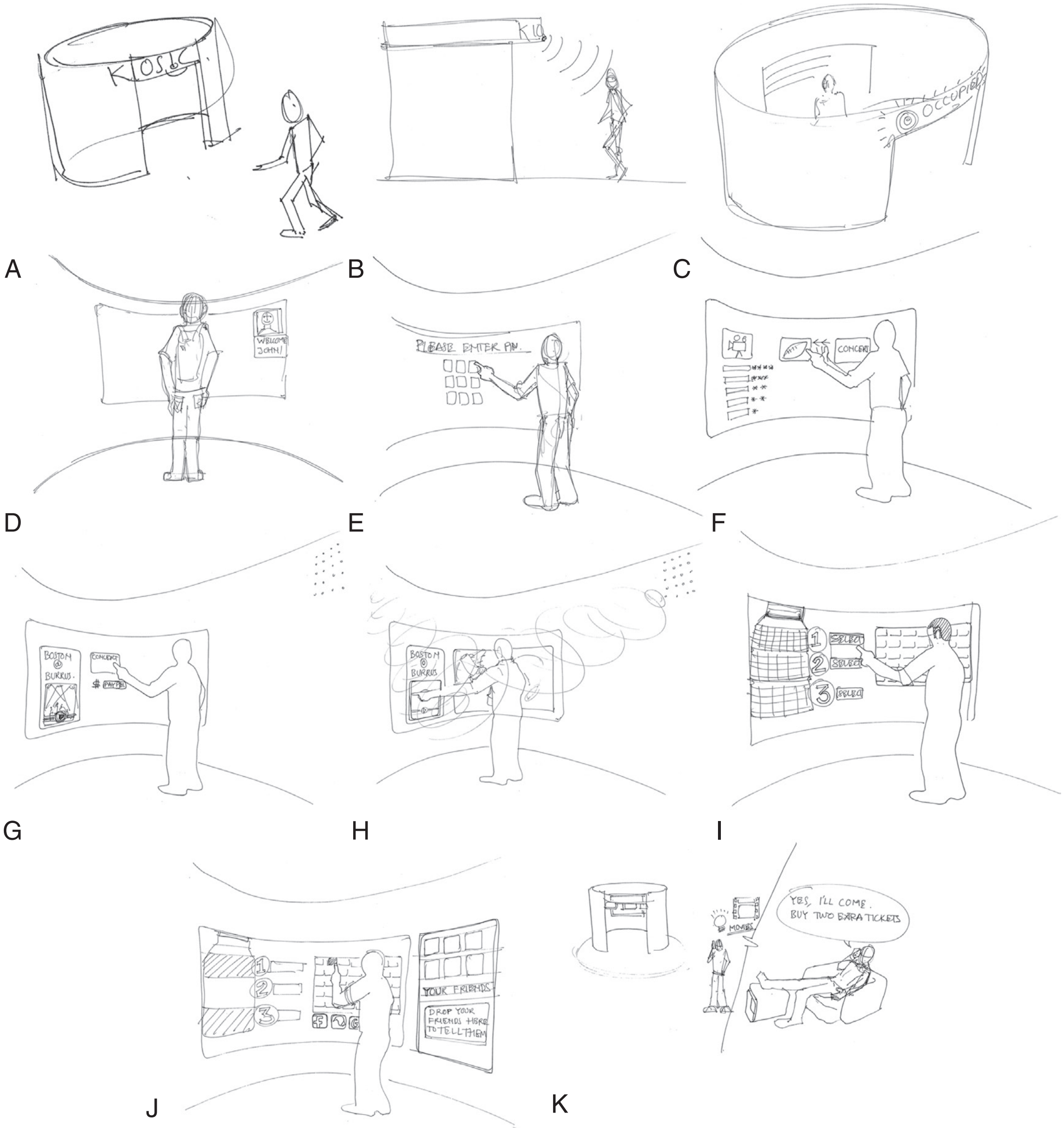

Fig. 16.4

Sample sketches for a concert ticket purchase storyboard (from Akshay Sharma, Virginia Tech Department of Industrial Design.).

Example: Ticket Kiosk System Design Scenario

A corresponding design scenario may resemble the following:

- Ticket buyer approaches the kiosk.
- Sensor detects user presence and begins the immersion protocol.
- Activates "Occupied" sign on the wraparound case.
- Detects people with Middleburg University (MU) passports.
- Center screen: Greets buyer and asks for PIN.
- Center screen: Shows recommendations and most popular current offering based on buyer's category.
- Screen on right: Shows buyer's profile if one exists on MU system.
- Screen on left: Lists options, such as browse events, buy tickets, and search.
- Center screen: Buyer selects "Boston Symphony at Burruss Hall" from the recommendations.
- Screen on right: "Boston Symphony at Burruss Hall" title, information, and images.
- Surround sound: Plays music from that symphony.
- Center screen: Shows "pick date and time."
- Center screen: Buyer selects date from the month view of the calendar (can be changed to week view).
- Screen on right: The entire context selected so far, including date.
- Center screen: A day view with times, such as matinee or evening. The rest of the slots in the day show related events, such as wine tasting or special dinner events.
- Screen on left: Options for making reservations at these special events.
- Center screen: Buyer selects a time.
- Center screen: Available seating chart with names for sections/categories and aggregate number of available seats per each section.
- Screen on left: Categories of tickets and prices.
- Center screen: Buyer selects category/section.
- Screen on right: Updates context with time and selected category/section.
- Center screen: Immerses user from a perspective of that section. Expands that section to show individual available seats. Has a call to action "Click on open seats to select" and an option to specify number of seats.
- Screen on left: Options to go back to see all sections or exit.
- Center screen: Buyer selects one or more seats by touching on available slots.
- Center screen: Shows payment options and a virtual representation of selected tickets.
- Screen on left: Provides options with discounts, coupons, sign up for mailing lists, etc.
- Center screen: Buyer selects a payment option.
- Center screen: Provided with a prompt to put credit card in slot.
- Center screen: Animates to show a representation of the card on screen.

- Center screen: Buyer completes payment.
- Screen on left: Options for related events, happy hour and dinner reservations, etc. These are contextualized to the event for which they just bought tickets.
- Center screen: Animates with tickets and credit card coming back out of their respective slots.

This example is a good illustration of the breadth we intend for the scope of interaction, including a person walking in proximity to the kiosk, radio-frequency identification at a distance, and audio sounds being made and heard. This scenario can be used directly in the creation of wireframes.

16.5 EXAMPLES OF USER STORIES FOR THE INTERACTION PERSPECTIVE

Example: Ticket Kiosk System User Stories

Using the format suggested in Section 10.2.2.1, the following are examples of possible relevant user stories:

As a TKS ticket buyer,
I want to be able to hear sample sound bites from available concerts
So that I can be better informed about selecting a concert to attend.

As a TKS ticket buyer,
I want to be able to see visual options for available seating,
So that I can find what I think are the best seats for an event.
As a TKS ticket buyer,
I want privacy,
To protect me from others seeing my transactions.

As a TKS ticket buyer,
I want to be able to pay with a credit card, debit card, or Pay Pal,
So my payment will be safe and convenient.

Can you see how these example user stories could influence scenarios as inputs to the design?

16.6 WIREFRAMES

A wireframe is a line drawing representation of one state (one interaction screen) within an interaction design. The lines, outlines, and boxes of wireframe sketches have the appearance of a wireframe. Because a wireframe is a kind of prototype, the major place we describe wireframes is in the prototyping chapter (see Section 18.3). However, because a wireframe is also an interaction design representation, we need to introduce wireframes here.

16.6.1 What Are Wireframes?

Wireframes comprise lines and outlines (hence the name) of boxes and other shapes to represent emerging interaction designs. Wireframes are:

- Schematic diagrams and sketches that define a webpage or screen layout and navigational flow
- Used to illustrate high-level concepts, approximate visual layout, and behavior
- Sometimes used to show look and feel for an interaction design
- Embodiments of screen or other state transitions during usage, depicting envisioned task flows in terms of user actions on user interface objects

UX designers can create design representations quickly and inexpensively by sketching simple boxes, lines, and other shapes that can be labeled, moved, and resized. Text and graphics representing content and data in the design are placed in those objects. Drawing templates are used to provide quick representations of the more common kinds of user interface objects (more on this in the following sections). These wireframes are best generated with a software tool (such as Figma[1]).

In the early stages of design, typical wireframes are deliberately unfinished looking and usually contain little visual content (e.g., graphics, colors, font choices).

Fig. 16.5 provides a sketch of a generic wireframe to highlight types of active objects that can be used in wireframe designs.

16.6.2 The Path to Wireframes

Fig. 16.6 shows the path from ideation and sketching, task interaction models, and envisioned design scenarios to wireframes as representations of your designs for screen layout and navigational flow.

Along with ideation, sketching, and critiquing, task interaction models and design scenarios are the principal inputs to storytelling and communication of designs. As sequences of sketches, storyboards are a natural extension

[1]https://www.figma.com/

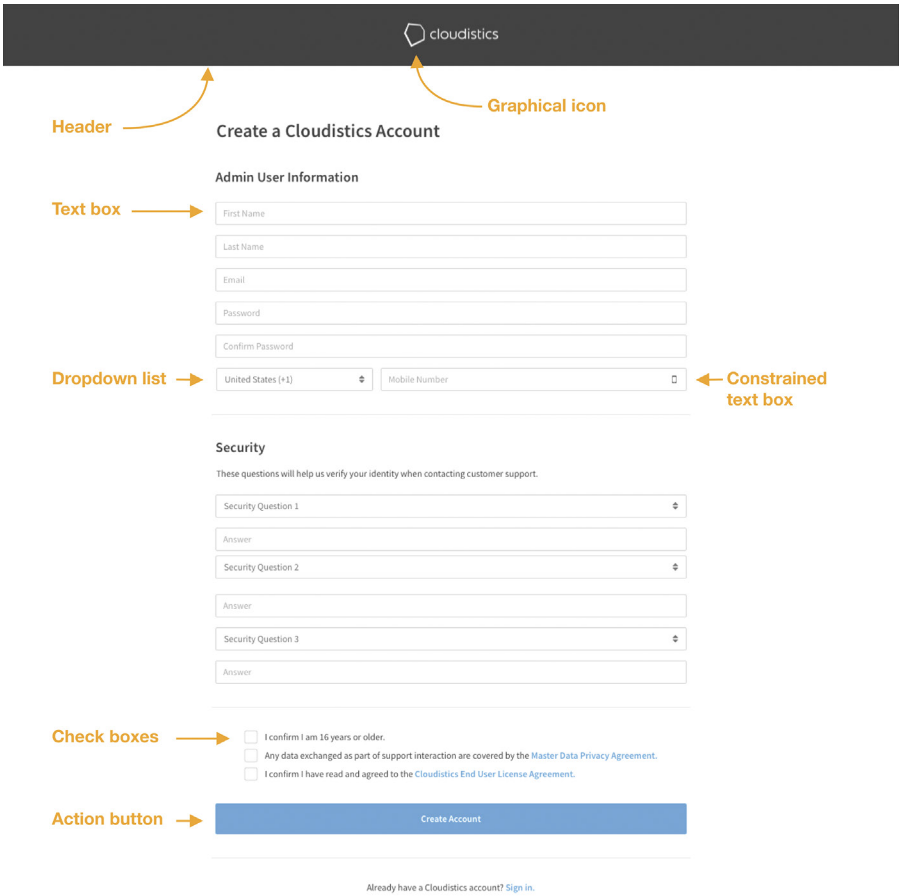

Fig. 16.5

Example generic wireframe with design elements.

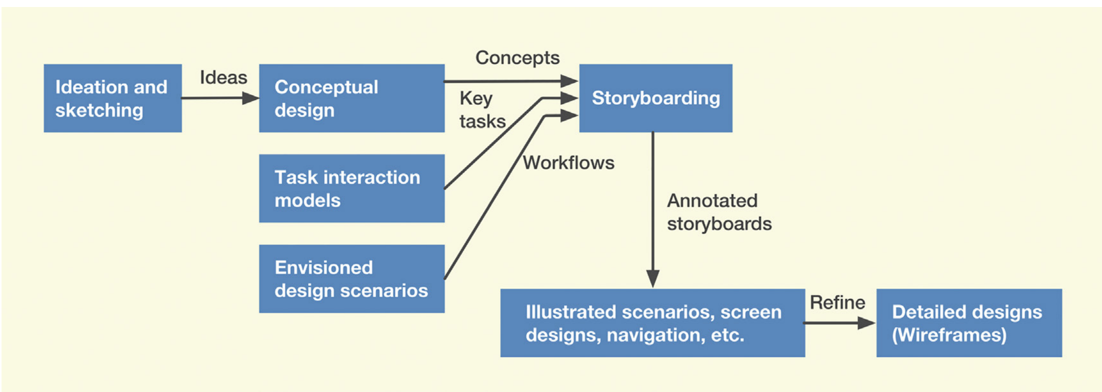

Fig. 16.6

The path from ideation and sketching, task interaction models, and envisioned design scenarios to wireframes.

of sketching. Storyboards, like scenarios, typically represent only selected task threads. Fortunately, it is a short and natural step from storyboards to wireframes.

To be sure, nothing beats pencil/pen and paper or a whiteboard for the sketching needed in ideation (brainstorming for design creation [see Chapter 13]), but at some point, when the design concept emerges from ideation, it must be communicated to others who pursue the rest of the lifecycle process. Wireframes have long been the choice in the field for documenting, communicating, and prototyping these interaction designs.

16.6.3 Wireframe Design Elements

Low-fidelity wireframes usually do not have graphical design elements (e.g., images, specific colors, typography). Typical elements represented in a wireframe can include the following:

- Header
- Footer
- Content areas
- Labeling
- Menus
- Tabs (possibly with dropdowns)
- Buttons
- Icons
- Popups
- Messages
- Navigation bar, navigation links
- Placeholders for logo and branding images
- Input fields (for search, etc.)

16.7 INTERMEDIATE INTERACTION DESIGN

As the team continues to refine the interaction design, the workflows explored using high-level storyboards are put to critique and analysis. This results in any necessary adjustments and whittling down of ideas under consideration. The team then focuses on the most promising ideas by increasing the fidelity of the designs and fleshing out more details of the flows, working toward a wireframe prototype.

16.7.1 Introducing a National Park Website Example

As a step toward creating a wireframe prototype, we introduce an example involving a website for a fictitious national park, where users can browse attractions and outdoor activities and can reserve park resources, such as campsites. We begin with examples of early intermediate interaction design for this product on a smartphone device.

At this point, it will be helpful to establish a few work-domain notes for camping in national parks.

- National parks can be very large.
- One national park can contain more than one campground.
- Many campgrounds will have their own websites.
- Sometimes larger campgrounds will be divided into areas or sectors.
- For a given campground, a campsite reservation will have several parameters.

16.7.1.1 Example: National park reservation website user stories

Let us begin with a few examples of user stories for the campground reservation website:

As a campground website user seeking a campsite reservation,
I want to be able to find a campground near a location that I want to travel to,
So that I can decide my destination location first and then find a nearby campground.

As a campground website user seeking a campsite reservation,
I want to be able to search for campsites available on a desired date(s)
So that I can match up site availability with my own needs.

As a campground website user seeking a campsite reservation,
I want to be able to find campsites that offer the facilities I want or need (such as electric power, water, and sewer service),
So that I can have the most comfortable camping experience offered.

As a campground website user seeking a campsite reservation,
I want to be able to arrange payment via credit card, debit card, Pay Pal, or other method,
So that payment will be safe and convenient.

16.7.1.2 Example: A persona for Annie, a user of a national park campground reservation website

Name

Annie Simpson[2]

Age

28 years old

Marital status

Single

Location

Boulder, CO

Job title

Dog groomer

Bio

Annie owns a small dog grooming business. She loves the outdoors and has a deep emotional attachment to being in nature. She is relatively experienced in camping with her small pull-behind camper trailer. She has a few part-time helpers who take care of the business when she is away.

Goals

She loves dog grooming, but also hopes to be a writer someday, writing non-fiction accounts of her travels and camping in the West.

Frustrations

She has to use computers, but mainly to keep track of her dog clients. And she does some online shopping. When she does use a computer, especially on a website, she doesn't like complicated options for things she doesn't want, anyway. Her needs are simple, and she wants her interaction to be simple and easy to understand, too. Otherwise, she doesn't like high-tech applications, preferring what she calls the "reality" of nature.

Technical abilities

Annie is not computer-savvy. She is a visually oriented person, so she prefers to navigate and make online choices visually rather than having to wade through text.

[2]Photo courtesy Dr. Lana Beex, Amsterdam

We add a little narrative here to support the connection between Annie and our campground reservation system design. Annie loves camping and the outdoors but not technology and computers. She will have little patience with complicated task structures and sequences on websites. She will not like having her time taken up wrestling with options not used. Rather, she likes reliable (consistent) and easy-to-understand interaction with no frills and few complex choices.

She needs to easily locate a national park campground that is reasonably nearby. Her preference to operate visually means it is better if she can select a campground visually from a map rather than from a list of names. The same applies to selecting a campsite within a campground—better for Annie to do it visually from a campground map than from a list.

Exercise 16.2 Creating a User Persona for Your Product or System

Goal: Get some experience at writing a persona (see Section 5.6.1.2).
Activities: Select an important work role within your system. At least one user class for this work role must be very broad, with the user population coming from a large and diverse group, such as the general public.

- Using your user-related UX research data, create a persona, give it a name, and get a photo to go with it.
- Write the text for the persona description.

Deliverables: One- or two-page persona write-up.
Schedule: You should be able to complete this in 1 hour.

16.7.1.3 Example: A design scenario for a national park campground reservation system

Can you see where the description of the user's journey in this scenario has been influenced by the user stories of Section 16.7.1.1 and the persona for Annie of Section 16.7.1.2?

Annie is planning a camping trip to visit the state of Utah. Here is a terse description of Annie's journey as a kind of work activity scenario. After considering state parks and private campgrounds, Annie has decided to camp in her small camper trailer at a national park. She searches for national parks in and near Utah, looking online at park attractions and features, and decides on Bruce Valley National Park. Then she performs the following actions:

1. Visit Bruce Valley National Park home screen.
2. Search for campgrounds in or near this park.

3. Find a map of the Bruce Valley National Park area, overlaid with icons for campgrounds. Notice how a visual map of the campground matches Annie's visually oriented persona (Section 16.7.1.2).
4. Select a promising campground and see pop-up, named the John Muir Campground.
5. Click icon and go to John Muir Campground page.
6. Follow link to see reviews of the campground.
7. Satisfied, go back to John Muir Campground page and follow link to "Camping" and another to "Check campsite availability" (Fig. 16.7). From this figure, we show early-stage sketches of intermediate design wireframes depicting basic workflow for reserving a campsite (simplified for illustration).

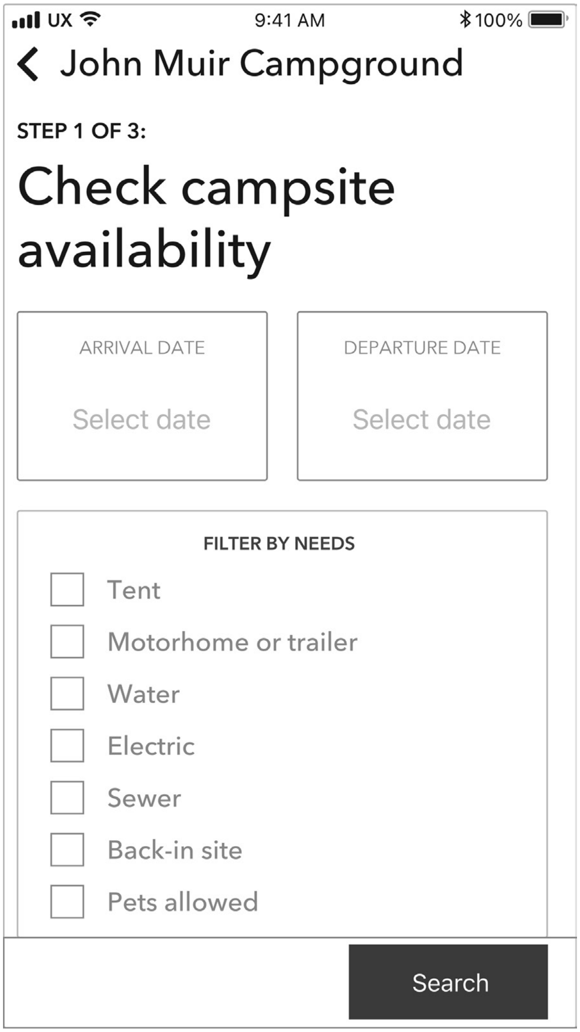

Fig. 16.7
Screen for search to check campsite availability.

8. Enter reservation information, including arrival date and departure date, on the pop-up calendar widget.
9. Select preferences in search filter (e.g., motorhome, water, electric, sewer). Notice how the option for campsites where pets are allowed in Fig. 16.7 matches the love of dogs in Annie's persona (Section 16.7.1.2).
10. Click "Search" to find site availability.
11. Site availability results shown (Fig. 16.8) by color highlighting (blue) of available site icons on a campground map, the default view. In this case, site A03 is shown as selected from among the available sites.
12. If the user prefers to see available sites in a list format, clicking "List" on the right-hand side near the top gets the list displayed (Fig. 16.9).

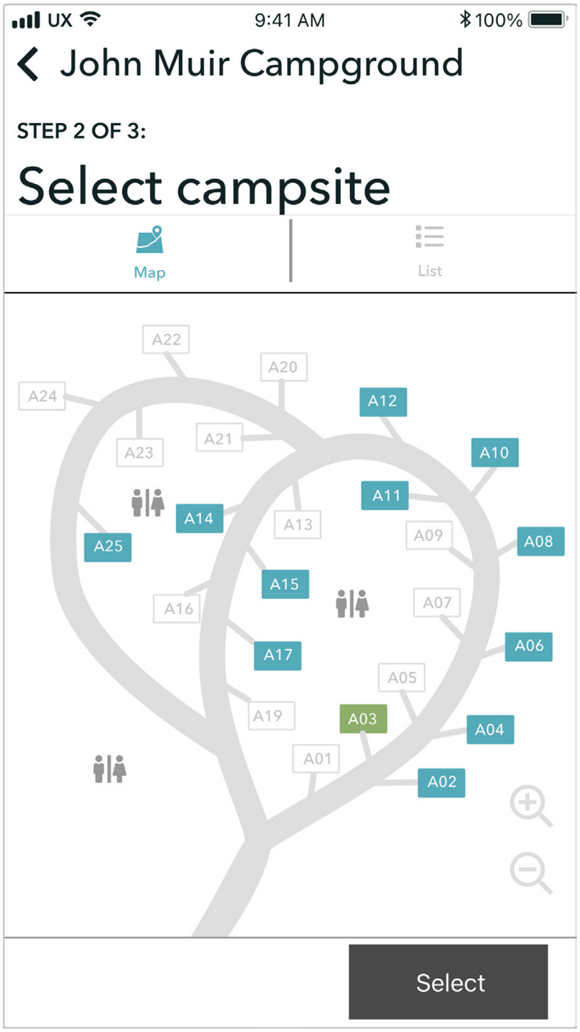

Fig. 16.8

Highlighted site icon in campground site map showing site A03 is selected from the available sites.

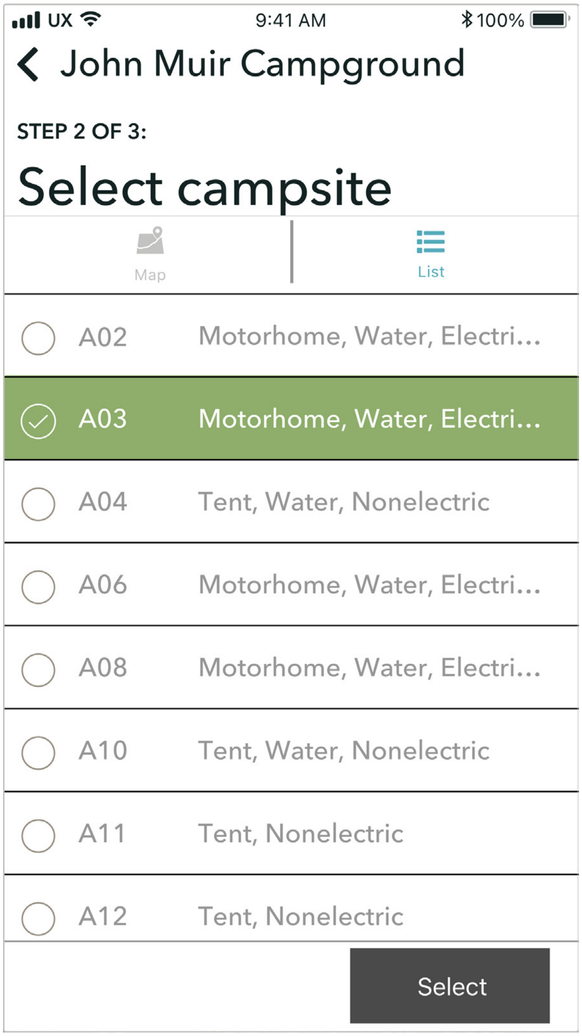

Fig. 16.9
Site availability is displayed in an alternative list form.

13. Select the A03 site icon from the screen of Fig. 16.8 or click the A03 selection circle in Fig. 16.9 and click "Select" to express choice of this site, as summarized in Fig. 16.10.

14. If the details of selected campsite look good, click "Reserve" (see Fig. 16.10) to make the reservation and proceed to the "Make payment" page (Fig. 16.11).

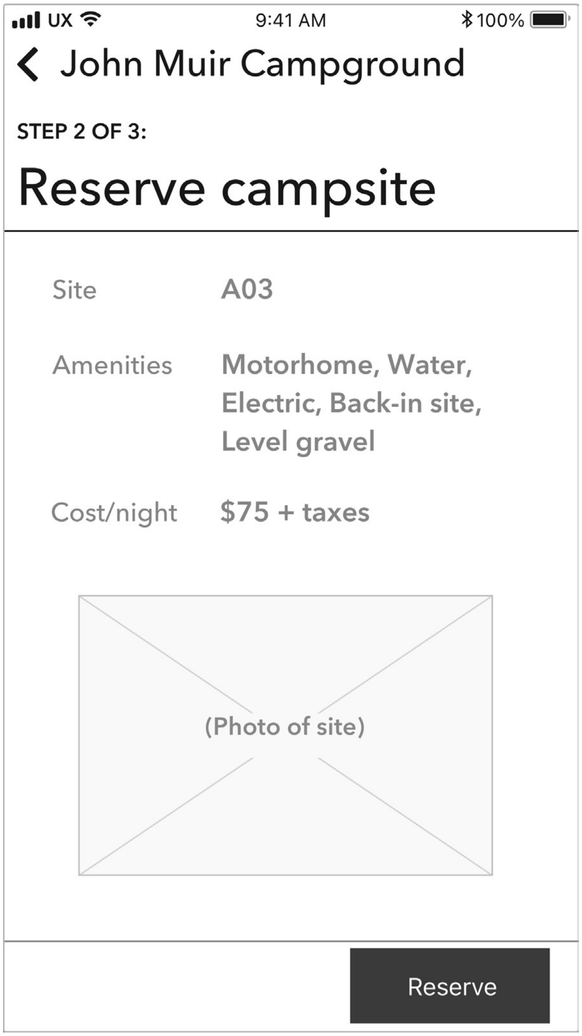

Fig. 16.10
"Reserve campsite" page.

16.7.1.4 Feedback from client and users

These early wireframes raised questions when subjected to critiquing by the client and some users experienced in this work domain. Asking and answering questions like these is an important part of the journey toward high-fidelity design.

1. Are the criteria used to filter a list of campsites in Fig. 16.7 the right set to use? For example, the choice of "Electric" hookup at a site needs additional details on whether it is 30A or 50A. It was recommended that we involve larger numbers of experienced campers in vetting this part of the design.

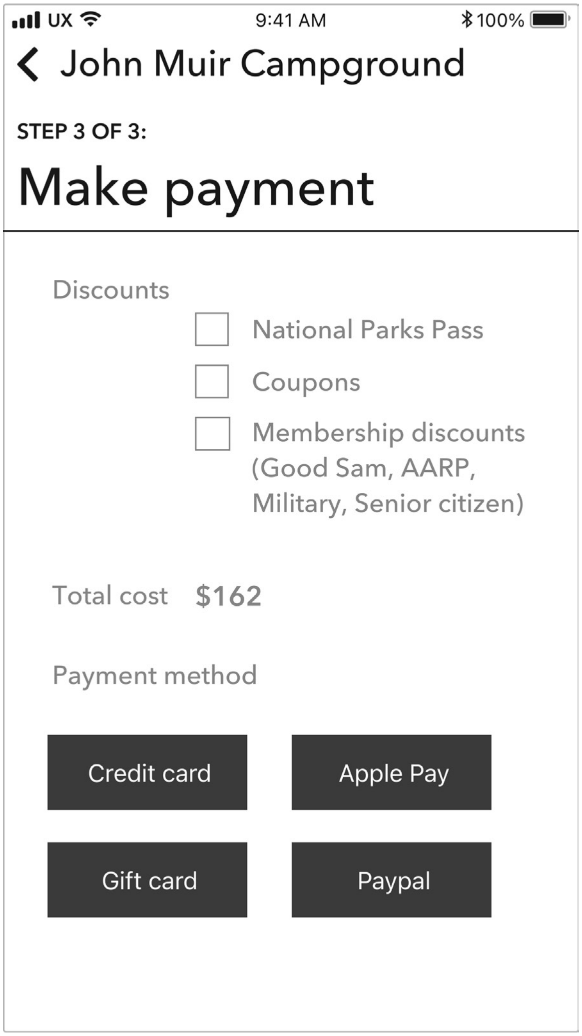

Fig. 16.11
Screen for "Make payment" page.

2. How will the interaction in Fig. 16.8 work if the campsites on the map are too close to one another? Will the users have enough of a tap area to select one versus another (especially on a smartphone)? Or can the user enlarge the view to separate close campsites? How can the design help the user make a choice of campsites?
3. In addition to the views in Figs. 16.8 and 16.9, users asked for a list view that not only gave availability for the dates requested but showed availability, by site number, for days before and after the requested dates, seen by moving a slider to scroll the dates. This is to serve the needs of users who do not find availability for the requested dates but are flexible and could use other nearby dates. This had to be tabled for future

consideration because of the complexity of the software needed to support this functionality.

4. How much information can be provided in the description portion of the campsite preview in Fig. 16.10? For example, current design includes just one picture of the site on the description page. Will that be enough for the user to decide on that site?
5. For a confirmation screen (not shown) following Fig. 16.11, at the end of a reservation transaction, it was suggested to ask if there are any further reservations needed. Sometimes one user wants to reserve adjacent sites for themselves and friends, and this feature would obviate the need to go back through the whole reservation sequence while risking someone else getting the adjacent site(s) in the meantime. This change is quite easy to design and implement.

16.7.1.5 Increasing presentation fidelity to visualize the final product experience

As the design team answers these questions through iteration, they slowly start filling in more details for the design. If the underlying design concept holds up to critiquing via these iterations, the team ends up with a final interaction design candidate for that device.

As an example of how full-fidelity interaction design along the presentation dimension can be, Fig. 16.12 shows an example of high-fidelity wireframes in an alternate visual design of envisioned flow.

Fig. 16.13 shows additional high-fidelity wireframe mockups of imagined workflow for evaluating look and feel.

16.8 INTERACTION DESIGN PRODUCTION

The final candidates for interaction are prototyped and subjected to evaluation (see Parts 4 and 5). Any issues identified in the evaluation kick off another round of design modifications and prototyping. Once the design is finalized, it is described in detail in a phase called design production. The objective of this phase is to define the design in enough detail that it becomes a specification for software engineers to implement.

Fig. 16.14 shows the detailed design specification for a park page of the desktop device interaction design. Note the annotations in the bottom defining details, such as how many lines the preview of the description should be and what happens when a user hovers over certain elements on the screen.

VIEW A CAMPING AREA

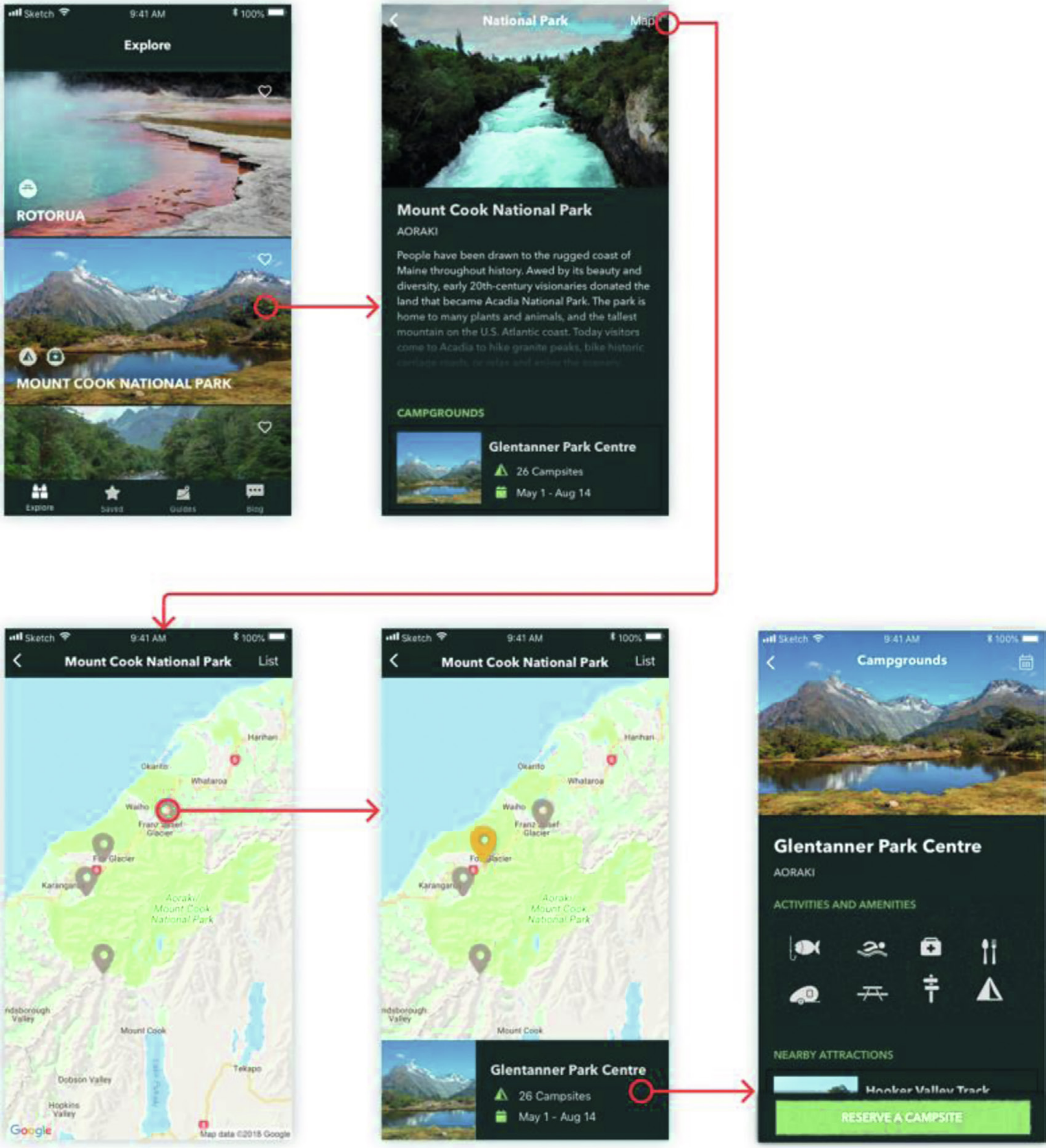

Fig. 16.12

As an example of attractive full-fidelity graphics, the high-fidelity wireframes show the look and feel of an imagined workflow (from Ame Wongsa, Director of UX, Fungible, Inc.; Christina Janczak, Product Design Manager, Arista Networks, Inc.).

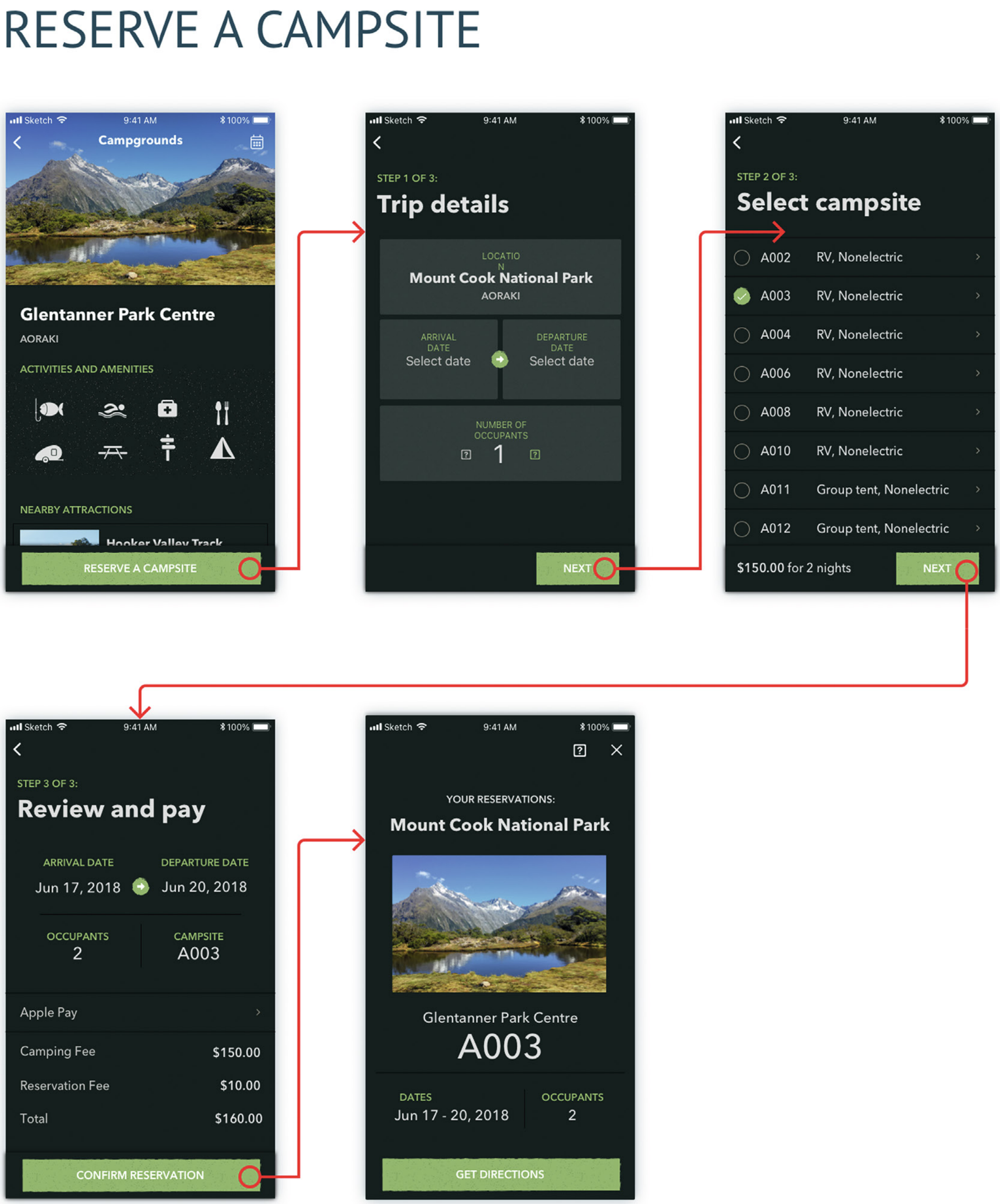

Fig. 16.13

Additional high-fidelity wireframes that were not accepted (from Ame Wongsa, Director of UX, Fungible, Inc.; Christina Janczak, Product Design Manager, Arista Networks, Inc.).

Design notes that go with Fig. 16.14 include the following:

- Descriptions will show six lines of text. If the description is longer, use truncation with "More…". On click, expand to show the full description and push down the next section.
- List all campgrounds for the national park. When the user mouses-over an entry in the campgrounds list, show a marker indicating the location on the mini-map.
- The mini-map shows the immediate location. The map sticks to the top of the view when the screen scrolls. A marker indicates the location of a highlighted campground.

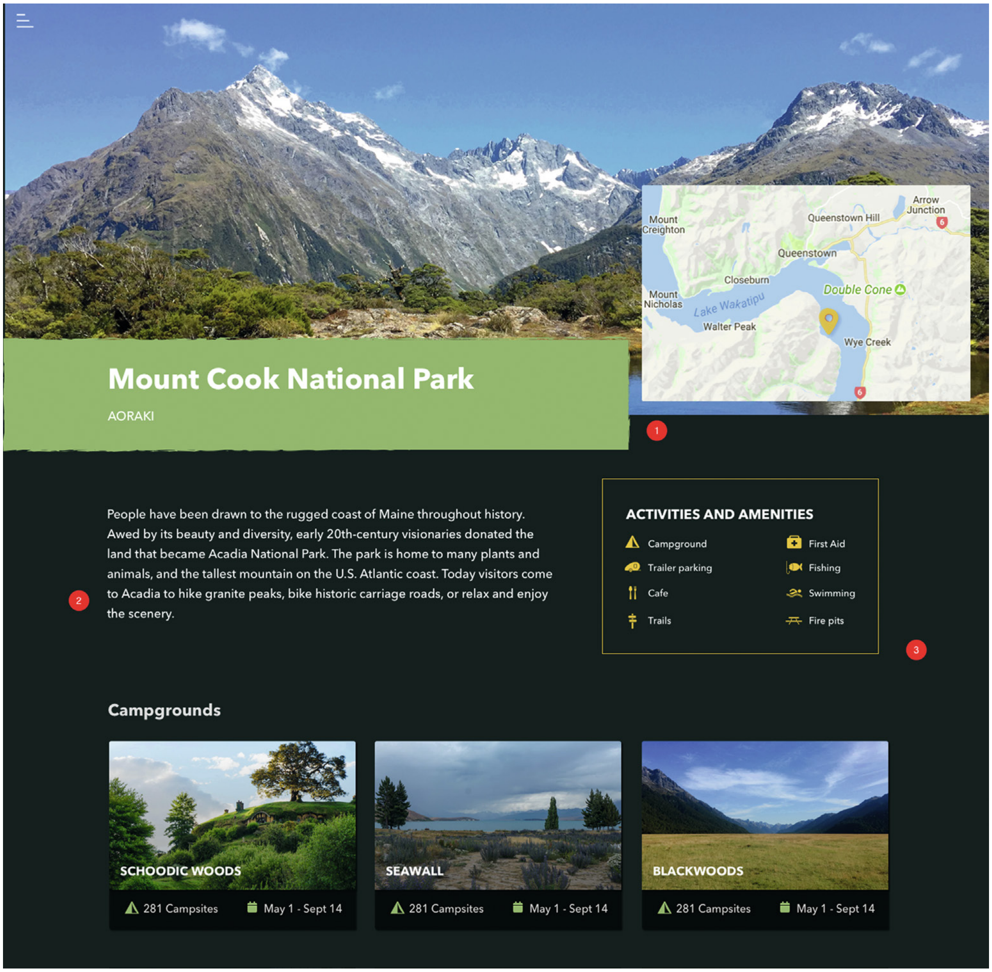

Fig. 16.14

Detailed graphical design for a park page in the desktop device view (from Ame Wongsa, Director of UX, Fungible, Inc.; Christina Janczak, Product Design Manager, Arista Networks, Inc.).

Exercise 16.3 Intermediate and Detailed Design for Your Product or System

Goal: Practice developing a few parts of the intermediate and detailed design (see Section 5.6.1.2).

Activities: If you are working with a team, get together.

- Choose one principal work role for your system (e.g., the customer).
- Choose one key task that work role is expected to perform.
- For that work role and task, make a few illustrated scenarios to show associated interaction.
- Sketch some annotated screen layouts (wireframes) to support your scenarios, along with representation of the navigational structure.
- Go for a little depth, but not much breadth.

Hints, cautions, and assumptions

- Do not get too involved in design details yet (e.g., icon appearance or menu placement).
- Base your screen designs on the UX research and design you have done so far.

Deliverables: Just the work products that naturally result from these activities.

Schedule: Whatever you can afford. At least give it an honest try.

16.9 ADOPT AND MAINTAIN A DESIGN SYSTEM

As you get into detailed interaction design, it is important to maintain consistency of design vocabulary and styles within and across the various devices. A design system is the way to ensure such consistency.

16.9.1 What Is a Design System?

A design system is a repository that is fashioned and maintained by designers to capture and describe details of visual and other general design decisions, especially about screen designs, patterns, user interface widgets, font choices, iconography, and color usage, which can be applied in multiple places. Its contents can be specific to one project or an umbrella guide across all projects on a given platform or over a whole organization. A design system offers a standardized vocabulary of interaction elements and syntax that helps with consistency and reuse of design decisions. It also affords efficient generation of designs by pre-making (and specifying) lower-level design choices. Every organization needs one.

Because your design decisions continue to be made throughout the project and because you sometimes change your mind about design decisions, the design system is a living repository that grows and is refined along with the design. Typically, this repository is private to the project team and is used only internally within the development organization.

16.9.2 Why Use a Design System?

Following are reasons for designers to use a design system within a project:

- It helps with project control and communication. Without documentation of the large numbers of design decisions, projects—especially large projects—get out of control. Everyone invents and introduces their own design patterns, possibly different each day. The result almost inevitably is poor design and a maintenance nightmare.
- It is a reliable force toward design consistency. An effective design system helps reduce variations of the details of widget design, layout, formatting, color choices, and so on, giving you consistency of details throughout a product and across product lines.
- A design system is a productivity booster through reuse of well-considered design patterns. It helps avoid the waste of reinvention.

16.9.3 What To Put in a Design System?

Your design system should include all the kinds of user interface objects and design situations where your organization cares the most about consistency (Meads, 2010). Most design systems are very detailed, spelling out the parameters of graphic layouts and grids, including the size, location, and spacing of user interface elements. This includes widget (e.g., buttons, dialogue boxes, menus, message windows, toolbars) usage, position, and design. Also important are the layouts of forms, including the fields, their formatting, and their location.

Your design system is the appropriate place to standardize fonts, color schemes, background graphics, and other common design elements. Other elements of a design system include interaction procedures, interaction styles, message and dialogue fonts, text styles and tone, labeling standards, vocabulary control for terminology and message and label wording, and schemes for deciding how to use defaults and what defaults to use. It should be worded very specifically, and you should spell out interpretations and conditions of applicability.

You should include example design guidelines and pictures taken from designs to make it communicate visually. Supplement with clear explanatory text. Incorporate examples of incorrect usage of elements, including specific examples of UX problems found in evaluation related to design system violations.

Your design system is also an excellent place to catalog design patterns (Borchers, 2001); your standard ways of constructing and placing menus, buttons, icons, and dialogue boxes; and your use of color for user interface objects, such as buttons. Perhaps among the most important parts of a design system are rules for organizational branding.

We will revisit design systems in Section 18.3.2, in the context of prototyping, to talk about how modern wireframing tools help achieve consistency and efficiency with minimal effort.

16.10 SEGUE

We have covered designing for ecological needs in the previous chapter and interaction needs in this chapter. We will wrap up Part 3 by tackling designing for emotional needs in the next chapter.